

YOUR LOGO HERE	Statistical Data Analysis Report Study: LoB and LoD	
	Assay name: SimulacioDadesLoBLoD Platform: Platform Project code: IA	Edition 1
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1 Introduction

This is the report of the Limit of the Blank (LoB) and Limit of Detection (LoD) for the SimulacioDadesLoBLoD assay for lot Lot1 in the Platform platform. The report has been automatically generated. Data analysis has been executed by Pol Monte. The LoB and LoD studies were performed following CLSI document EP17-A2, Evaluation of Detection Capability for Clinical Laboratory Measurement Procedures; 2nd Edition.

LoB determination: Each of 4 blank samples were tested with 5 replicates for 3 days.
LoD determination: Each 4 Low-level samples were tested with 5 replicates for 3 days.

1.1 Study Acceptance Criteria

LoB does not have any Specification or Acceptance Criteria. LoD does not have Acceptance Criteria. LoD will be considered acceptable if they meet the specification shown in Table 1.

Table 1. LoD Acceptance Criteria and Specifications

Acceptance Criteria	Specification (mg/m3)
Not Applicable	< 45.00

2 Materials and Equipment Description

The materials and equipment used in this LoB and LoD studies are described in Section 5 of the corresponding DRC-248-01.

3 Exploratory Data Analysis

The exploratory data analysis helps to have a first impression about the data. It includes Summary results for LoB and LoD studies.

3.1 LoB Summary Results

Table 2 contains the statistical summary with the mean and standard deviation (SD) for blank samples data.

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Table 2. Blank Samples Statistical Summary

Sample	N	Mean (mg/m3)	SD (mg/m3)
LoB1	15	0.65	1.84
LoB2	15	0.70	1.67
LoB3	15	0.39	1.52
LoB4	15	0.50	1.13

Abbreviations: SD, Standard Deviation.

Figure 1 is a histogram with the distribution of the frequencies of all blank measurements.

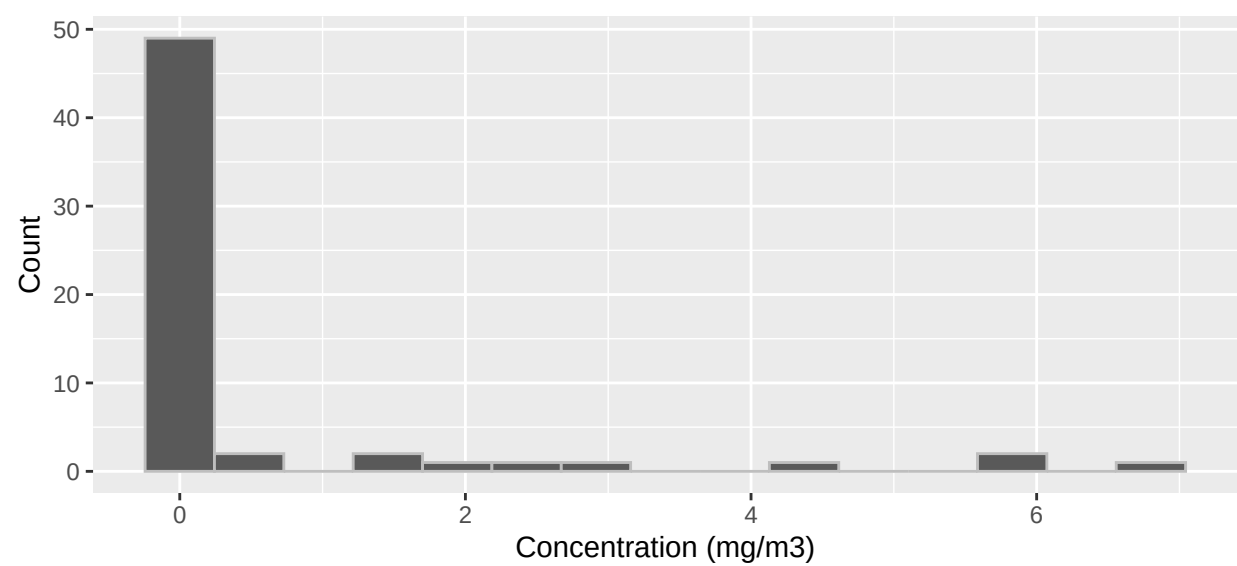


Figure 1. LoB Histogram

3.2 LoD Summary Results

Table 3 contains the statistical summary with the mean and standard deviation (SD) for low-level samples data.

Table 3. Low-level Samples Statistical Summary

Sample	N	Mean (mg/m3)	SD (mg/m3)
LoD1	15	45.09	17.69
LoD2	15	47.07	18.00
LoD3	15	47.93	18.82
LoD4	15	50.00	18.61

Abbreviations: N, Sample Size; SD, Standard Deviation.

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3.3 LoD Normality Test

Shapiro-Wilk test was used for analyzing if the LoD data follows a normal distribution. These results were used to know if for LoD study is necessary to perform a data analysis using a parametric or a Non-parametric analysis.

If p-value is ≥ 0.05 , the distribution of the data is not significantly different from a normal distribution and therefore LoD can be calculated using a parametric approach. Whereas if p-value is <0.05 the data doesn't follow a normal distribution and therefore LoD is recommended to be calculated using a Non-parametric approach.

Figure 2 shows Normality Q-Q Plot, where curve red lines indicate the 95% of confidence interval.

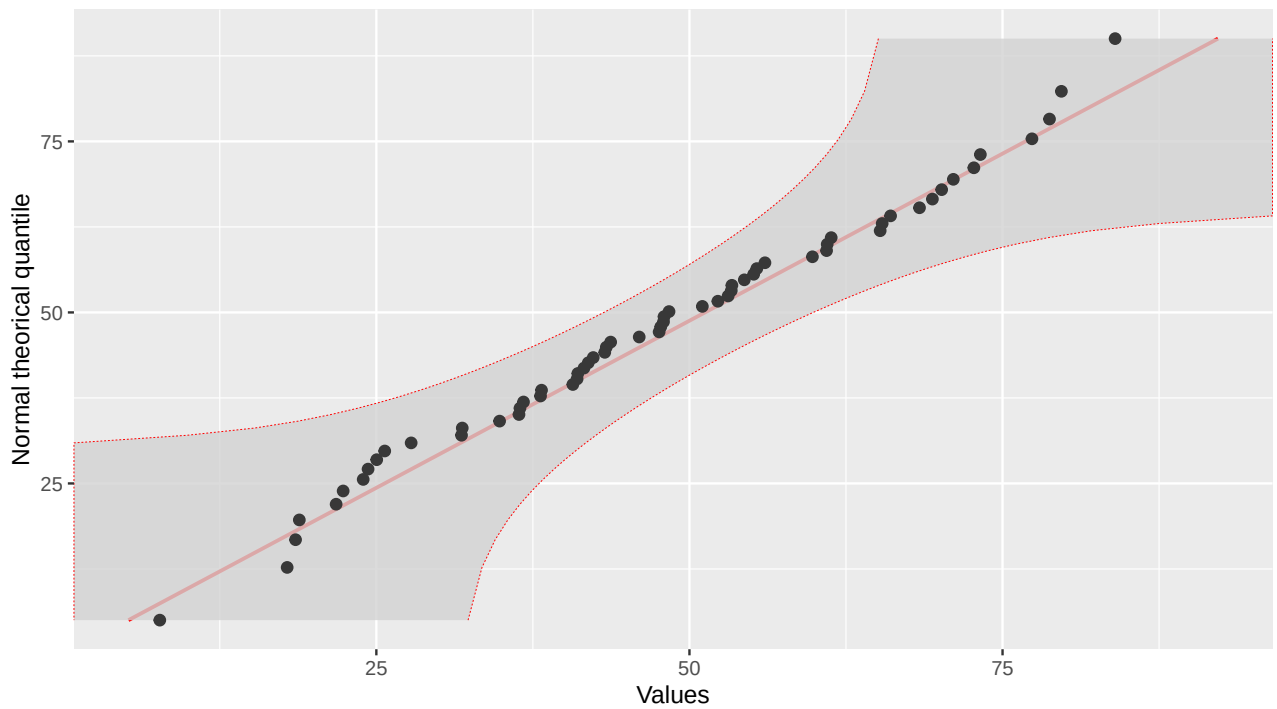


Figure 2. Normality Q-Q Plot

The p-value obtained using Shapiro-Wilk test is 0.68.

4 Results

4.1 LoB Analysis Result

LoB was calculated by sorting all blank samples measurement results ($n = 5$) from lowest to highest for all blank samples tested (total of 4). Then LoB was calculated using Rank Position formula referenced in CLSI EP17 2nd Edition, using the typical Type I error risk of $\alpha = 0.05$, where the corresponding percentile for the distribution of blank sample

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results (PctB) = 0.95.
The rank position corresponding to this PctB was computed as:

$$Rank\ Position = 0.5 + (BPct_B)$$

where B is the number of blank sample measurements.

The calculated rank position is 57.5.

Table 4 shows the Rank Positions from blank sample test Results.

Table 4. Rank Positions and LoB Results

Rank position	Value (mg/m3)
55	2.50
56	2.75
57*	4.20
58*	5.90
59	6.00
60	6.80

* Indicates the rank positions used to calculate LoB

The Limit of Blank is 5.05 mg/m3.

Figure 3 is a histogram with the distribution of the frequencies of all blank measurements. A vertical red line indicates where the LoB is situated.

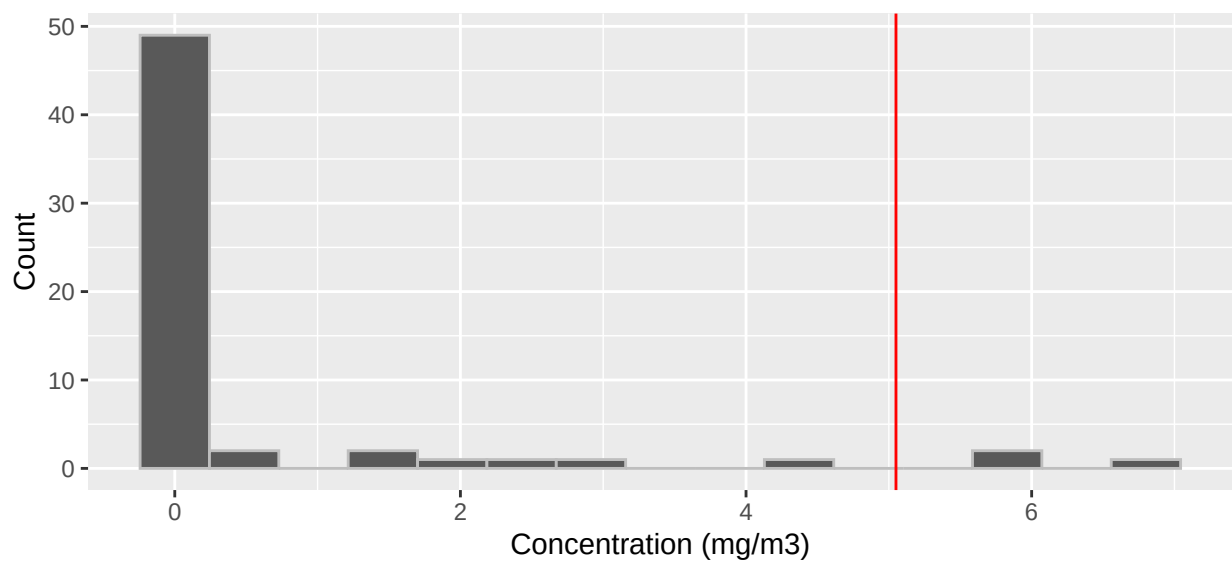


Figure 3. LoB Histogram with LoB reference line

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4.2 LoD Parametric Analysis Result

LoD was calculated using the Parametric Analysis. The pooled standard deviation (SDL) and the multiplier to give the 95th percentile of a normal distribution (Cp) coefficient were calculated.

The LoD estimate for each reagent lot was calculated per equation references in the CLSI EP17 2nd Edition by classical approach, using the reported LoB determined above.

$$LoD = LoB + c_pSD_L$$
$$c_p = \frac{1.645}{1 - (\frac{1}{4(L-J)})}$$

where Cp is a multiplier to give the 95th percentile of a normal distribution, L = total number of all low level sample results across all reagent lots, J = number of low level samples.

Table 5 shows the LoD results obtained with Parametric Analysis.

Table 5. LoD Parametric Analysis Result

SDL (mg/m3)	cp	LoD (mg/m3)	Specification (mg/m3)	Evaluation
18.29	1.65	35.26	< 45.00	PASS

The Limit of Detection (LoD) is 35.26 mg/m3.

4.3 LoD Non-parametric Analysis Result

LoD was also calculated using the Non-parametric Analysis.

All measurement results across the low-level samples for a given reagent lot were combined into a single distribution, and then the percentage of individual results falling below the LoB value was calculated. If the percentage is less than 5% (desired Type II (β) error (0.05), the LoD for that reagent lot may be taken as the median of the combined low-level sample result distribution. In some cases, in order to non-overestimate and correct the shift created by the median of LoD samples being too high, an adaptation of the non-parametric formula presented in the CLSI EP17-A2 was used. The formula subtracts the 5th percentile and then it adds the calculated LoB:

$$LoD = LoB + median - 5^{th}percentile$$

If the percentage of low-level samples below LoB is higher than 5% the study should be repeated. Therefore the evaluation should be considered not applicable (N/A).

Table 6 shows the LoD results obtained with Non-parametric analysis.

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Table 6. LoD Non-parametric Analysis Result

% under LoB	Median (mg/m3)	LoD (mg/m3)	Specification (mg/m3)	Evaluation
0.00	47.64	34.04	< 45.00	PASS

The Limit of Detection (LoD) is 34.04 mg/m3.

4.4 Results Summary

Considering the results obtained in Table 4 The Limit of Blank for the SimulacioDadesLoBLoD assay on the platform Platform with lot Lot1 is 5.05 mg/m3.

If it is needed to test two or three reagent lots, then the LoB of the measurement procedure is a maximal value of the LoBs obtained for each reagent lot.

Considering the results obtained in Table 5 the Limit of Deteciton (LoD) with parametric analysis for the SimulacioDadesLoBLoD assay on the platform Platform with lot Lot1 is 35.26 mg/m3. The results meet specifications.

Considering the results obtained in Table 6 the Limit of Deteciton (LoD) with Non-parametric analysis for the SimulacioDadesLoBLoD assay on the platform Platform with lot Lot1 is 34.04 mg/m3. The results meet specifications.

If there are 2 or 3 reagent lots, then the LoD of the measurement procedure is a maximal value of the LoDs obtained for each reagent lot.

The p-value from Shapiro-wilk test is ≥ 0.05 , LoD parametric test is recommended.

5 Software

This report has been generated using the following software:

- IVDReport version 1.0.0
- R version 4.2.2 (2022-10-31 ucrt)
- tidyverse package version 2.0.0
- reshape2 package version 1.4.4
- qqplotr package version 0.0.6
- zip package version 2.3.0
- knitr package version 1.44
- kableExtra package version 1.3.4

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6 Operator and Data Input Reviewer

Operators:

- Pol (25/06/2024)
- Marc (03/07/2024)
- Ian (27/06/2024)
- Xavier (05/07/2024)

Data Input Reviewers:

- Marina (28/06/2024)
- Gerard (03/08/2024)
- Ignasi (30/062024)
- Pere (07/07/2024)

7 Summary Raw Data

Data entry was performed by two different data input reviewers via two different Excel files. Previous to the data analysis, application automatically verified that data in both Excel files were identical.

Table 7. Summary of the Raw Data used in the LoB analysis

Day	Sample	Rep1 (mg/m3)	Rep2 (mg/m3)	Rep3 (mg/m3)	Rep4 (mg/m3)	Rep5 (mg/m3)
1	LoB1	0.00	0.00	0.00	0.00	0.25
2	LoB1	2.75	0.00	6.80	0.00	0.00
3	LoB1	0.00	0.00	0.00	0.00	0.00
1	LoB2	0.00	0.00	0.00	6.00	0.00
2	LoB2	2.50	0.00	0.00	0.00	0.00
3	LoB2	0.00	0.00	2.00	0.00	0.00
1	LoB3	0.00	0.00	0.00	0.00	0.00
2	LoB3	0.00	5.90	0.00	0.00	0.00
3	LoB3	0.00	0.00	0.00	0.00	0.00
1	LoB4	0.00	0.00	0.00	0.00	4.20
2	LoB4	0.50	0.00	1.50	0.00	0.00
3	LoB4	0.00	0.00	0.00	1.25	0.00

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Table 8. Summary of the Raw Data used in the LoD analysis

Day	Sample	Rep1 (mg/m3)	Rep2 (mg/m3)	Rep3 (mg/m3)	Rep4 (mg/m3)	Rep5 (mg/m3)
1	LoD1	21.79	83.99	47.93	43.24	54.38
2	LoD1	47.58	42.32	73.23	40.69	18.55
3	LoD1	52.27	38.11	34.84	24.34	53.09
1	LoD2	43.36	59.82	41.92	31.80	77.35
2	LoD2	41.03	70.14	18.85	36.39	41.09
3	LoD2	38.19	79.70	47.97	53.34	25.03
1	LoD3	31.86	61.00	65.39	65.23	17.88
2	LoD3	78.75	61.32	43.71	56.03	27.78
3	LoD3	25.67	45.99	23.95	48.37	66.06
1	LoD4	36.47	41.58	55.13	36.75	47.71
2	LoD4	69.40	71.07	60.95	22.35	53.39
3	LoD4	72.71	68.37	51.03	7.71	55.38